10.7 Circles in the Coordinate Plane

Name _____ Period ____ Date _____

Geometry 10.7 · Student sheet

I can…

- write the equation of a circle from its center and radius.
- find the center and radius of a circle from its equation and graph it.
- decide whether a point is on, inside, or outside a circle.

Vocabulary · fill in the missing word

Standard equation of a circle — $(x - h)^2 + (y - k)^2 = r^2$, where (h, k) is the _____ and r is the radius

Center — the point inside a circle that is the same _____ from every point on the circle

Radius — a _____ from the center to a point on the circle (or its length)

Part 1 · Discovery: points 5 away

Plot each point and find its distance from (0, 0). Draw the circle with center (0, 0) through the points that are 5 away.

Point	$x^2 + y^2$	Distance from (0, 0)	On the circle?
(3, 4)			
(−4, 3)			
(0, −5)			
(2, 4)			

Every point on the circle fits the equation _____.

Part 2 · Notes and examples (fill in each note, then work the example)

Standard equation of a circle: $(x - h)^2 + (y - k)^2 = r^2$, with center _____ and radius _____. With the center at the origin: $x^2 + y^2 = r^2$.

Example 1

Write the standard equation of each circle.

a. center (−2, 5), radius 3

b. center (0, 0), radius 4

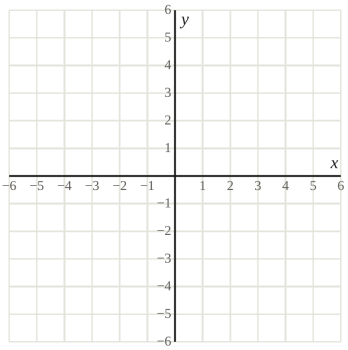
From the center and a point: find the radius with the _____ formula, then write the equation.

Example 2

A circle has center (1, −3) and passes through (4, 1). Write its standard equation.

Example 3

Find the center and radius of the circle $(x - 1)^2 + (y + 2)^2 = 9$. Then graph it.



Not in standard form: _____ for x and for y, then read the center and radius.

Example 4

Find the center and radius of the circle $x^2 + y^2 - 6x + 4y - 3 = 0$.

On, inside, or outside: compare the point's distance from the center with the _____.

Example 5

A cell tower at $(-1, 2)$ reaches every phone within 5 miles. Write an equation for the edge of its range. Is a phone at $(3, 4)$ in range?

Part 3 · Partner practice

Partner 1

Write the standard equation of the circle with center $(4, -1)$ and radius 6.

Partner 2

Find the center and radius of the circle $x^2 + y^2 + 8x - 2y + 8 = 0$.